

Module 6

Code with GitHub Codespaces



Agenda



GitHub Foundations Part 1

- 1 Introduction to Git
- 2 Introduction to GitHub
- 3 Introduction to GitHub's products
- 4 Configure code scanning on GitHub
- 5 Introduction to GitHub Copilot
- 6 Code with GitHub Codespaces
- 7 Manage your work with GitHub Projects
- 8 Communicate effectively on GitHub using Markdown

GitHub Foundations Part 2

- 9 Contribute to an open-source project on GitHub
- 10 Manage an InnerSource program by using GitHub
- 11 Maintain a secure repository by using GitHub best practices
- 12 Introduction to GitHub administration
- 13 Authenticate and authorize user identities on GitHub
- 14 Manage repository changes by using pull requests on GitHub
- 15 Search and organize repository history by using GitHub
- 16 Using GitHub Copilot with Python

Introduction

VS Code yaml
 json

GitHub Codespaces is an instant, cloud-based development environment that uses a container to provide you with common languages, tools, and utilities for development.

By the end of this module, you'll:

- Explore the Codespaces lifecycle and processes.
- Review the ways you can customize your Codespace set up.
- Compare the differences between GitHub Codespaces and GitHub.dev.
- Complete an exercise to practice coding in Codespaces.

The Codespace lifecycle

There are four ways to create a Codespace:



- From a GitHub template GitHub.com to start a new project.
- From a branch in your repository.
- From an open pull request, to explore work-in-progress.
- From a commit in a repository's history to investigate a bug.

Code editor

VS Code Desktop
Local installation

VS Code Browser
Chrome, Safari, Edge

JetBrains IDE
IntelliJ IDEA, GoLand, RubyMine, etc.

Command line
SSH via GitHub CLI

→
Your editor is how you view and edit your codespace

←
Changes to your codespace are reflected back in your editor

Azure
Hosting

Virtual Machine
Linux hardware

Docker
Container
Docker development environment

A clone of your repository
Source code

Languages
Python, Ruby, etc

Tooling
Extensions, linting, etc

CPUs

Disk storage

Codespace

Personalize your Codespace

Settings Sync and Dotfiles

- Synchronize VS Code settings between the desktop application and the web client.
- Use a dotfiles repository to specify scripts, shell preferences, and other configurations.

Codespace Customization

- Rename Codespaces to differentiate between them.
- Change the shell in a Codespace, open new terminal windows with a preferred shell, and configure shells using dotfiles.

Machine Type and Editor Settings

- Change the machine type running your Codespace to match your resource needs.
- Set your default editor for Codespaces in your personal settings, with options including VS Code, JetBrains Gateway, and JupyterLab.

Region and Timeout Settings

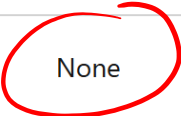
- Set your default region in the GitHub Codespaces profile settings to personalize data location.
- Specify a default timeout period for Codespaces, with a default of 30 minutes of inactivity.

Codespaces versus GitHub.dev editor



Use GitHub.dev to navigate files and sources code repositories from GitHub, and make and commit code changes.

Open any repository, fork, or pull request in GitHub.dev editor.

	GitHub.dev	GitHub Codespaces
Cost	Free 	Free monthly quota of usage for personal accounts.
Availability	Available to everyone on GitHub.com	Available to everyone on GitHub.com.
Startup	GitHub.dev opens instantly with a key-press and you can start using it right away without having to wait for configuration or installation.	When you create or resume a Codespace, the Codespace is assigned a VM. The container is then configured based on the contents of a devcontainer.json file. This setup takes a few minutes to create the development environment.
Compute	There are no associated compute resources, so you can't build and run your code or use the integrated terminal.	With GitHub Codespaces, you get the power of a dedicated VM to run and debug your application.
Terminal access	None 	GitHub Codespaces provides a common set of tools by default, meaning that you can use the Terminal exactly as you would in your local environment.
Extensions	Only a subset of extensions that can run on the web appear in the extensions view and can be installed	With GitHub Codespaces, you can use most extensions from the Visual Studio Code Marketplace.

Exercise - Code with Codespaces and Visual Studio Code



Now that you understand the Codespaces lifecycle and processes, it's time to practice coding in Codespaces and Visual Studio Code (VS Code).

This exercise walks you through the following steps:

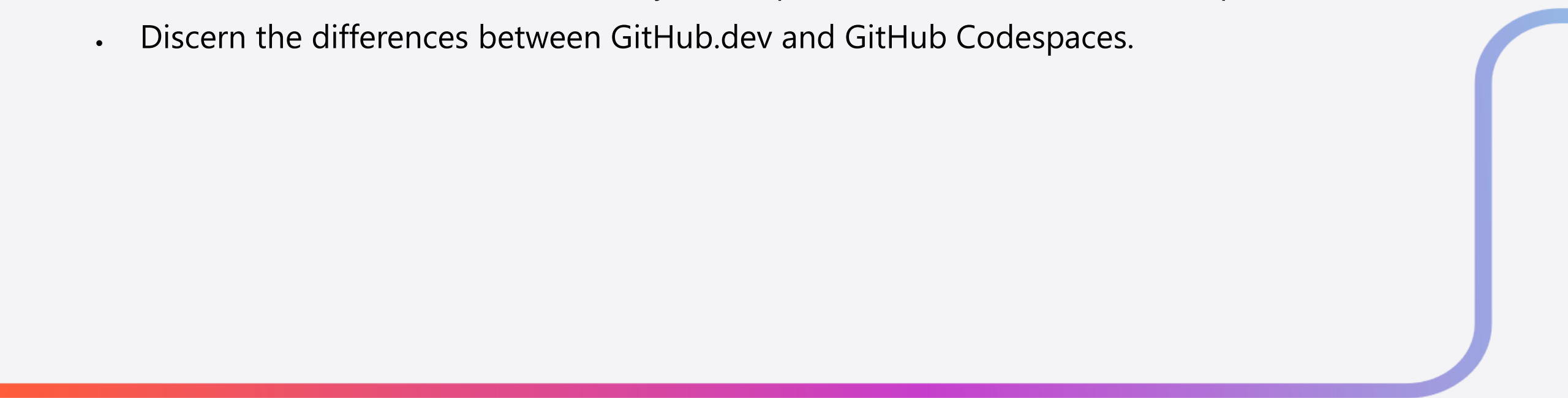
- Create your first codespace
- Add a custom image
- Customize your codespace
- Personalize your codespace



Start course

Summary

Now that you've finished this module, you should be able to:

- Describe GitHub Codespaces.
 - Explain the GitHub Codespace lifecycle and how to perform each step.
 - Define the different customizations you can personalize with GitHub Codespaces.
 - Discern the differences between GitHub.dev and GitHub Codespaces.
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End of presentation



